

Young Tableau Again

Combinatorics

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1 Revision on Notations

Notation	Meaning
$\lambda = (\lambda_1, \lambda_2, \dots, \lambda_k)$	λ is a partition of n if $\lambda_1 + \dots + \lambda_k = n$ and
λ'	the dual partition of λ (Think about the Young diagram) e.g. the dual partition of $(5, 4, 4, 1)$ is $(4, 3, 3, 3, 1)$
λ_i	The i -th term in the partition λ (= number of boxes in i -th row of Young diagram)
λ'_i	The i -th term in the dual partition of λ (= number of boxes in i -th column of Young diagram)
f^λ	Number of Standard Young Tableaux of the shape λ
$h(i, j)$	The hook length of box (i, j) (i.e. number of boxes in the hook of (i, j))
$\text{dg}(\lambda)$	the Young diagram of shape λ (or more specifically, the set of boxes of it)
$\text{rw}(T)$	the reading word of Standard Young Tableau T
$P(w)$	the Young Tableau generated by the word w by Row Bumping Algorithm

2 Probabilistic Proof for Hook Length Formula

Recall the Hook Length Formula, which tells us how many Standard Young Tableaux there are of a given shape λ :

Theorem 2.1 (Hook length formula) Let λ be a partition of n

$$f^\lambda = \frac{n!}{\prod_{(i,j) \in \text{dg}(\lambda)} h(i, j)}$$

where the sum is taken over all boxes of the Young diagram of shape λ

Before we begin, we need the following simple lemma. A **corner box** of λ is a box which has no boxes above or on the right of it (i.e. it is on an upper-right corner of the Young diagram/Tableau) (again we are using the French notation, instead of the English one)

Lemma 2.2 Let (r, s) be a corner box of a Young Diagram of shape λ , and (i, j) be any box in it. Then $h(i, j) = h(r, j) + h(i, s) - 1$

In the following we define an algorithm which generates a Standard Young Tableau of a given shape λ . We hope that each of the Tableau appears with the same probability, but, a priori, there seem to be no simple way to do so without knowing the whole set of Standard Young Tableaux of shape λ .

Algorithm 2.3 (Tableau Generation Algorithm)

Input: shape λ , size n

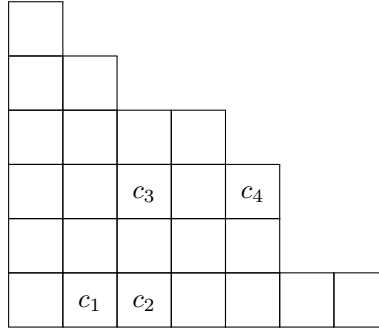
Steps:

1. Randomly choose a box c from the Young diagram $\text{dg}(\lambda)$
2. While $h(c) > 1$ (i.e. c is not a corner cell)
 - (a) Randomly choose a cell c' in the hook of c (each cell is chosen with a probability of $\frac{1}{h(c) - 1}$)
 - (b) Replace c by c' , go back to step 2.
3. Now c is a corner box, we write the number n in the box c . To determine the numbers in the remaining boxes, repeat from step 1, but with the new input of: the Young Diagram $\text{dg}(\lambda) - \{c\}$ and size $n - 1$

Check that this algorithm must terminate (since $h(c)$ is a strictly decreasing sequence of positive integers), and it will results with a Standard Young Tableau

Let the **hook walk** $(c_1, c_2, \dots, c_k)$ be the sequence of boxes that c has ever become, in time order. Write $c_s = (i_s, j_s)$. Define the **row set** $I = \{i_1, \dots, i_{k-1}\} - \{i_k\}$ be the rows that the hook walk (excluding the row of the last, corner cell) has ever walked on. Similarly define **column set** as $J = \{j_1, \dots, j_{k-1}\} - \{j_k\}$

Example 2.4 Suppose the input are $n = 24, \lambda = (7, 5, 5, 4, 2, 1)$, a possible outcome is:



The row set is $I = \{1\}$ and the column set is $J = \{2, 3\}$. Note that we have $|I| + |J| = n - 1$. The probability of this hook walk to be choosen is:

$$\frac{1}{n} \cdot \frac{1}{h(c_1) - 1} \cdot \frac{1}{h(c_2) - 1} \cdot \frac{1}{h(c_3) - 1} = \frac{1}{24 \cdot 9 \cdot 7 \cdot 3}$$

We will show the following lemmas:

Lemma 2.5 Given a partition λ of n , corner box $c = (r, s)$, and row and column sets $I \subset \{1, 2, \dots, r-1\}, J \subset \{1, 2, \dots, s-1\}$, the probability that the hook walk in the Young Diagram of shape λ that ends at corner box c , and has row and column sets I, J such that is

$$p(\lambda, c, I, J) = \frac{1}{n} \prod_{i \in I} \frac{1}{h(i, s) - 1} \prod_{j \in J} \frac{1}{h(r, j) - 1}$$

proof induct on the number $|I| + |J|$, with the base case being $I = J = \phi$

Lemma 2.6 Given a partition λ of n and corner box $c = (r, s)$, the probability that a hook walk ends at c is

$$p(\lambda, c) = \frac{1}{n} \prod_{i=1}^{r-1} \frac{h(i, s)}{h(i, s) - 1} \prod_{j \in J} \frac{h(r, j)}{h(r, j) - 1}$$

proof this follows from Lemma 2.5, by summing over all the subsets I, J .

Theorem 2.7 Let λ be a partition of n , and S be a standard Young Tableau of shape λ , the probability that the Tableau Generation Algorithm with input λ, n gives S as output is

$$\frac{\prod_{(i,j) \in \text{dg}(\lambda)} h(i,j)}{n!}$$

proof Induction on n

Note that the Hook Length Formula follows directly from Theorem 2.7.

3 Knuth Equivalence and Increasing/Decreasing Subsequences

Definition 3.1 A word with letters $\{1, 2, \dots, n\}$ is a finite sequence of letters

Example 3.2 We can actually treat a permutation as a word, for example, the permutation $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 5 & 4 \end{pmatrix}$ corresponds to the word 23154.

Definition 3.3 Given a word w and a positive integer k , define $\text{inc}_k(w)$ as that, when we choose k disjoint increasing subsequences of w , the maximum number of letters that these k subsequences can possibly contain. Similarly define $\text{dec}_k(w)$

Exercise 3.4 Let $w = 134234122332$ be a word. Find $\text{inc}_1(w)$, $\text{inc}_2(w)$ and $\text{inc}_3(w)$.

It is in general quite difficult to find the numbers $\text{inc}_k(w)$, but it turns out that Young Tableau give us a simple way to calculate $\text{inc}_k(w)$. Actually our theory will work for any word, but for simplicity, we will only work on words with $1, 2, \dots, n$ each appearing exactly once (which correspond to Standard Young Tableau)

Definition 3.5 Given a Standard Young Tableau, the **reading word** of T is the word

$$\text{rw}(T) = T(k, 1)T(k, 2) \dots T(k, \lambda_k)T(k-1, 1)T(k-1, 2) \dots T(k-1, \lambda_{k-1}) \dots T(1, 1) \dots T(1, \lambda_1)$$

In human language, this means that we "read" the Young Tableau left to right, up to down (just like how we normally read) to produce a word.

Example 3.6 Given the Standard Young Tableau

8	9		
2	6	7	
1	3	4	5

Its reading word is 892671345.

Exercise 3.7 Given the word $w = 2617354$, find $P(w)$, $\text{rw}(P(w))$ and $P(\text{rw}(P(w)))$.

Fact 3.8

1. For a Standard Young Tableau T , $P(\text{rw}(T)) = T$
2. For a word w , $\text{rw}(P(w)) \neq w$ in general

Turns out different words can give the same Standard Young Tableau under Row Bumping Algorithm!
How do we solve this problem?

Definition 3.9 (Knuth Equivalence) The elementary Knuth relations are :

$$\begin{aligned} ayxz\mathbf{b} &\sim ayzx\mathbf{b} && \text{,for } \mathbf{a}, \mathbf{b} \text{ being words, and } x < y < z \\ axzy\mathbf{b} &\sim azxy\mathbf{b} && \text{,for } \mathbf{a}, \mathbf{b} \text{ being words, and } x < y < z \end{aligned}$$

Two words v, w are called **Knuth equivalent** (denoted by $v \equiv_K w$) if we can change v to w through a finite sequence of elementary Knuth relations.

To make it simple, we can swap two number xz if and only if there is a number y between x and z which lies immediately on left/right of xz .

Theorem 3.10 For any word v consisting of letters $1, 2, \dots, n$ each used once, we have $v \equiv_K \text{rw}(P(v))$

proof Note that for words $\mathbf{u}, \mathbf{v}, \mathbf{a}, \mathbf{b}$, we have $\mathbf{u} \equiv_K \mathbf{v}$ if and only if $\mathbf{aub} \equiv_K \mathbf{avb}$. Now we do induction on the length of v . (and to do this, we actually also do induction on l , the number of rows in the Young Tableau, so this is actually a double induction)

Theorem 3.11 Given words u, v , if $u \equiv_K v$, then $\text{inc}_k(u) = \text{inc}_k(v)$ and $\text{dec}_k(u) = \text{dec}_k(v)$

proof By induction, it suffice to deal with the case where u, v differ by one elementary Knuth relation. Let

$$u = ayxz\mathbf{b}, v = ayzx\mathbf{b}$$

where $\mathbf{a}, \mathbf{b}$ are words, and $x < y < z$. The following is a common trick in proving that two quantities (which are both maximum of some set) are equal: we prove the following two facts:

- If we have k disjoint increasing subsequences $\{I_1, I_2, \dots, I_k\}$ of u with a total of m letters, then we can find k disjoint increasing subsequences $\{I'_1, I'_2, \dots, I'_k\}$ of v also with a total of m letters (This shows that $\text{inc}_k(u) \leq \text{inc}_k(v)$)
- If we have k disjoint increasing subsequences $\{I_1, I_2, \dots, I_k\}$ of v with a total of m letters, then we can find k disjoint increasing subsequences $\{I'_1, I'_2, \dots, I'_k\}$ of u also with a total of m letters (This shows that $\text{inc}_k(u) \geq \text{inc}_k(v)$)

Theorem 3.12 (Our Main Theorem) Let w be a word with letters $1, 2, \dots, n$ each used once, and suppose $P(w)$ is a Standard Young Tableau with shape λ , then for all $k \geq 1$, we have

$$inc_k(w) = \lambda_1 + \lambda_2 + \dots + \lambda_k$$

$$dec_k(w) = \lambda'_1 + \lambda'_2 + \dots + \lambda'_k$$

Recall that λ_j is the number of boxes in the j -th row of the Young Tableau of shape λ , while λ'_j is the number of boxes in the j -th column of the Young Tableau of shape λ

proof Recall from Theorem 3.10 and 3.11, $w \equiv_K \text{rw}(P(w))$, so $inc_k(w) = inc_k(\text{rw}(P(w)))$, this means that we can WLOG assume that w is the reading word of some Standard Young Tableau. Now by a bit thinking we have the following facts:

- For an increasing subsequence of w , if we identify its letters in the Standard Young Tableau, then, in each column, there can be at most one letter contained in the increasing subsequence

Hence we can show the desired result

WARNING Be careful not to use Greedy Algorithm in your proof (recall from Example 3.4 that this might fail!)

Corollary 3.13 (Erdos-Szekeres Subsequence Theorem) Every word of length exceeding mn either has an increasing subsequence of length $m + 1$ or a decreasing subsequence of length $n + 1$.

And finally, here is another theorem that affirm the conjecture some of you might have been already be making in your mind, although this do not directly contribute to our work on increasing/decreasing subsequences.

Theorem 3.14 For any words u, v , $P(u) = P(v)$ if and only if $v \equiv_K w$

To prove this theorem, we need the following lemma:

Lemma 3.15

1. Suppose $v \equiv_K w$ are Knuth equivalent letters, suppose v', w' are obtained by erasing the largest letter n from v, w respectively. Then $v' \equiv_K w'$
2. Also, the Standard Young Tableau $T' = P(v')$ can be obtained from the Standard Young Tableau $T = P(v)$ by erasing the box containing the largest letter n

proof Exercise.